

Java Music Platform - Genius-Inspired Terminal App

This project is a console-based music platform written in Java, inspired by platforms like *Genius*. It allows users to interact with songs, albums, artists, lyrics, comments, Q&A, and more.

General Overview

The system has three types of users: **Admin**, **Artist**, and **User**. Each type has a specific set of features they can access:

1. **Admin**: Manages artist approvals and lyric edit requests.
2. **Artist**: Can create songs, albums, and request lyric edits.
3. **User**: Can interact with songs by liking/disliking, commenting, following artists, and viewing, Q&A.

The system stores information in .txt files for songs, artists, and user interactions. These files are saved in the data/ folder.

Workflow

1. **Login System**: When the program starts, the user is prompted to log in. Depending on the credentials entered, the system will route the user to the appropriate panel (Admin, Artist, or User).
2. **Admin Panel**: Admin has the ability to approve artist applications and handle lyric edit requests from users.
3. **Artist Panel**: Artists can create new songs or albums, and request lyric edits.
4. **User Panel**: Users can search songs, like/dislike songs, comment on songs, ask questions, answer questions, and follow artists.
5. **Lyric Edit Requests**: When a user or artist requests a lyric edit, it's stored and reviewed by either the artist or admin for approval. Once approved, the lyric file is updated.

6. **Song Creation and Storage:** Songs and albums are created by artists and stored as .txt files under the data/songs/ folder. Each file contains information such as song name, artist name, lyrics, likes, dislikes, and more.
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Class Overview

Login

- The Login class manages the login process and routes users to the correct panel based on their role (Admin, Artist, or User).
- The login system checks if the entered username and password match those stored in the system. If valid, it directs the user to their respective panel.

User

- The User class represents the regular user and provides a menu of options, including:
 - Searching for songs
 - Liking or disliking songs
 - Commenting on songs
 - Asking and answering questions about songs
 - Following artists
 - Viewing their following list
- The user can interact with songs through likes and dislikes, which are stored in dedicated files (data/likes/). Each like/dislike is recorded for a specific song and user.
- The user can also view a Q&A section where they can ask or answer questions related to specific songs.

Artist

- The Artist class allows artists to manage their content on the platform. The artist can:
 - Create new songs and albums, which are stored as .txt files in the data/songs/ directory.
 - Request lyric edits for their songs. The request is added to the pending list and can be reviewed by the artist or admin.
 - View pending lyric edit requests submitted by users.

Admin

- The Admin class is responsible for managing the platform. The admin has access to:
 - Approve or reject artist applications.
 - Review and approve/reject lyric edit requests from users.
- When an artist requests a lyric edit, the admin can approve it, and the changes are applied to the song's text file.

Song

- The Song class is the core class responsible for managing songs and albums. It handles:
 - Creating new songs and albums.
 - Displaying song details like name, artist, lyrics, and interaction data (likes, dislikes, views).
 - Handling lyric edit requests, which can be submitted by artists or users.
 - Storing song interactions like likes, dislikes, and comments.
 - Searching for songs based on keywords or random recommendations.
- Song data (lyrics, likes, comments, etc.) is stored in .txt files, and all user interactions (likes, dislikes, and comments) are saved in separate files under data/likes/.

Search

- The Search class allows users to search for songs by keyword. It checks both the song files and album files for matching content.
- It also provides a random song recommendation feature, where users can get a random song suggestion from the platform.

Question&Answer

- The Question&Answer class is responsible for managing the Q&A section of each song. It allows users to:
 - Ask questions about songs.
 - Answer existing questions.
- Questions and answers are stored within each song's data file, and they are displayed under the song lyrics.

Bonus Features

1. **File-based System:** All data (songs, likes, comments, Q&A, and lyric edits) is stored in plain text files. This makes it easy to update and manage data without requiring a database.
2. **Like/Dislike System:** Users can like or dislike songs, and this information is stored separately for each user and song in the data/likes/ folder.
3. **Q&A Section:** Each song has a Q&A section where users can ask questions and others can answer them. This adds an interactive layer to the song pages.
4. **Follow Artists:** Users can follow artists, and this data is stored in a file specific to each user (data/following_username.txt). Users can view and manage their following list.
5. **Random Song Recommendation:** The system provides a random song recommendation feature, where users can get a random song

suggestion from the entire platform, allowing them to discover new songs.

How to Run

1. Open the project in an IDE like IntelliJ or VS Code.
 2. Make sure the data/ folder exists, containing subdirectories for songs/, likes/, and other necessary files.
 3. Run the Login class to start the application.
 4. Follow the prompts to log in as an Admin, Artist, or User.
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This platform is designed for ease of use and extensibility. It's a simple, lightweight, and flexible implementation that could be expanded to include more features in the future.